CCNA LEARNING SERIES

Day 3: IPv4 Addressing & Subnetting

IPv4 • Binary basics • Network & Host portions • Subnet masks • CIDR • Practice

Learning objective
Understand how IPv4 addresses are structured, identify network and host portions, read subnet masks and CIDR notation, and calculate basic subnet details.

In this document

1. IPv4 address basics
2. Binary and octets
3. Network ID and Host ID
4. IPv4 address classes
5. Public and private IP addresses
6. Subnet mask and CIDR notation
7. Basic subnetting calculations
8. Troubleshooting commands
9. Interview questions and practice

A beginner-friendly reference guide with examples, revision notes and interview questions.

1. What is an IPv4 Address?

IPv4 stands for **Internet Protocol version 4**. It is a logical address used to identify a device on an IP network and help data reach the correct destination.

Example: 192.168.1.10

An IPv4 address contains 32 bits, divided into four groups called **octets**. Each octet contains 8 bits and is written as a decimal number from 0 to 255.

Octet 1	Octet 2	Octet 3	Octet 4
192	168	1	10
8 bits	8 bits	8 bits	8 bits

Remember: 4 octets × 8 bits = 32 bits. Each octet can have a value from 0 through 255.

2. Binary Basics

Computers work with binary digits: 0 and 1. Each 8-bit octet has place values 128, 64, 32, 16, 8, 4, 2 and 1.

Place value	128	64	32	16	8	4	2	1
Binary for decimal 192	1	1	0	0	0	0	0	0

192 = 128 + 64. Therefore, decimal 192 is 11000000 in 8-bit binary.

3. Network ID and Host ID

Network portion: identifies the network.
Host portion: identifies a device/interface within that network.

The subnet mask tells us which bits belong to the network portion and which belong to the host portion.

Example: 192.168.1.10/24
Subnet mask: 255.255.255.0
Network address: 192.168.1.0
Host address: 192.168.1.10
With /24, the first 24 bits are the network portion and the remaining 8 bits are the host portion.

4. IPv4 Address Classes (Traditional Classful Ranges)

IPv4 addresses were traditionally grouped into classes. Modern networks mainly use CIDR, but class ranges are still useful for basic networking knowledge.

Class	First-octet range	Traditional default mask	Typical use
A	1–126	255.0.0.0 (/8)	Large networks
B	128–191	255.255.0.0 (/16)	Medium networks
C	192–223	255.255.255.0 (/24)	Small networks
D	224–239	Not applicable	Multicast
E	240–255	Not applicable	Experimental / reserved

Note: 127.0.0.0/8 is reserved for loopback testing. Classful addressing is historical; CIDR is used for modern address allocation and routing.

5. Public and Private IPv4 Addresses

Private IP: used inside private networks such as homes, offices and labs. Private addresses are not routed directly across the public internet.
Public IP: globally unique address space used for internet communication, subject to allocation and routing.

Private range	CIDR	Address range
10.0.0.0 – 10.255.255.255	/8	10.0.0.0/8
172.16.0.0 – 172.31.255.255	/12	172.16.0.0/12
192.168.0.0 – 192.168.255.255	/16	192.168.0.0/16

Example: A home Wi-Fi router may assign a laptop 192.168.1.5 as a private IP address.

6. Subnet Mask

A subnet mask is a 32-bit value used with an IPv4 address to identify the network and host portions. Bits set to 1 represent the network portion; bits set to 0 represent the host portion.

CIDR	Subnet mask	Network bits	Host bits
/8	255.0.0.0	8	24
/16	255.255.0.0	16	16
/24	255.255.255.0	24	8
/25	255.255.255.128	25	7
/26	255.255.255.192	26	6
/27	255.255.255.224	27	5
/30	255.255.255.252	30	2

7. CIDR Notation

CIDR stands for **Classless Inter-Domain Routing**. The slash number shows how many bits are used for the network portion.

Example: 192.168.1.10/24 means 24 network bits and 8 host bits.

8. Basic Subnetting

Subnetting is the process of dividing one larger IP network into smaller logical networks called subnets. It helps organize networks and use IP address space efficiently.

Number of addresses per subnet = $2^{\text{host bits}}$
Traditional usable host addresses = $2^{\text{host bits}} - 2$

The usual usable-host formula excludes the network address and broadcast address. Special cases such as /31 point-to-point links and /32 host routes are handled differently.

Example 1: How many usable hosts in a /24 network?

A /24 has 32 – 24 = 8 host bits.
Total addresses = 2^8 = 256.
Usable host addresses = 256 – 2 = **254**.

Example 2: Divide 192.168.1.0/24 into /26 subnets

A /26 has 6 host bits, so each subnet has 2^6 = 64 total addresses and 62 traditionally usable host addresses. The block size is 64 in the last octet.

Subnet	Usable host range	Broadcast address
192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
192.168.1.64/26	192.168.1.65 – 192.168.1.126	192.168.1.127
192.168.1.128/26	192.168.1.129 – 192.168.1.190	192.168.1.191
192.168.1.192/26	192.168.1.193 – 192.168.1.254	192.168.1.255

Result: One /24 network is divided into four /26 subnets. Each /26 contains 64 total addresses and 62 traditionally usable host addresses.

9. Network Address, Usable Host Range and Broadcast Address

Network address: identifies the subnet itself; it is not assigned to a normal host.
Usable host range: addresses available for devices in the subnet.
Broadcast address: the last address in a traditional IPv4 subnet, used to reach all hosts in that subnet.

10. Useful Troubleshooting Commands

Command	Purpose
ipconfig	View IP configuration in Windows
ipconfig /all	View detailed IP configuration in Windows
ip addr	View IP addresses in Linux
ping 8.8.8.8	Test IP connectivity
tracert 8.8.8.8	Trace route in Windows
traceroute 8.8.8.8	Trace route in Linux (if installed)
ip route	View routing table in Linux
nslookup google.com	Check DNS name resolution

11. Interview Questions and Answers

Q1. What is IPv4?

IPv4 is Internet Protocol version 4. It is a 32-bit logical addressing system used to identify devices on IP networks.

Q2. How many bits are in an IPv4 address?

32 bits, divided into four 8-bit octets.

Q3. What is an octet?

An octet is a group of 8 bits in an IPv4 address.

Q4. What is a subnet mask?

A subnet mask identifies the network and host portions of an IPv4 address.

Q5. What does /24 mean?

It means 24 bits are used for the network portion and 8 bits are used for the host portion.

Q6. What is subnetting?

Subnetting divides a larger IP network into smaller logical networks called subnets.

Q7. How many usable hosts are in a /24 subnet?

There are 256 total addresses and traditionally 254 usable host addresses.

Q8. What is a network address?

It identifies the subnet and is not assigned to a normal host.

Q9. What is a broadcast address?

It is used to send traffic to all hosts on a particular IPv4 subnet.

Q10. Name the three private IPv4 ranges.

10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16.

Q11. What is CIDR?

Classless Inter-Domain Routing. Its slash notation indicates the number of network bits.

Q12. What is the difference between public and private IP addresses?

Private IPs are used inside private networks; public IPs are globally routable on the internet.

12. Practice Questions

Try these questions yourself before checking the answers.

1. How many bits are in one IPv4 octet?
2. What is the subnet mask for /24?
3. How many host bits are present in /26?
4. How many total addresses are in a /26 subnet?
5. How many traditionally usable host addresses are in a /26 subnet?
6. What is the network address of 192.168.1.70/26?
7. What is the broadcast address of 192.168.1.70/26?

Practice Answers

1. 8 bits
2. 255.255.255.0
3. 6 host bits
4. 64 total addresses
5. 62 usable host addresses
6. 192.168.1.64
7. 192.168.1.127

Quick Revision

Remember	Answer
IPv4 size	32 bits / 4 octets
Octet size	8 bits
Octet decimal range	0 to 255
/24 mask	255.255.255.0
/24 usable hosts	254
/26 total / usable	64 total / 62 usable
Private ranges	10/8, 172.16/12, 192.168/16

Study tip: Practise converting one octet between decimal and binary. Then calculate the network address, usable host range and broadcast address for a few /24, /26 and /27 examples.

End of Day 3 | Next topic: IPv4 Subnetting Practice & VLSM

Note: This guide covers introductory IPv4 addressing and traditional subnet calculations. Actual address planning can include special-purpose ranges and exceptions not covered in this beginner guide.